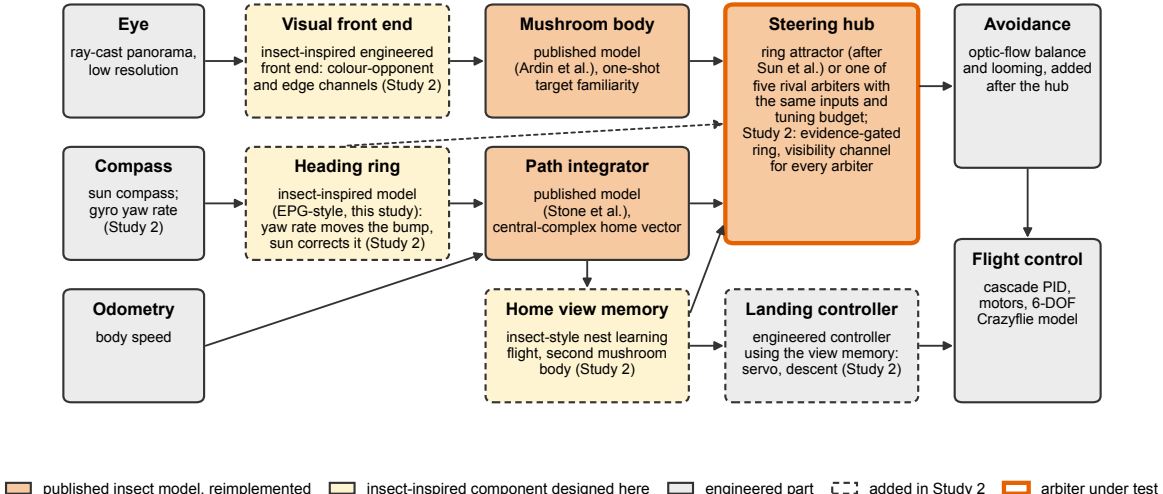
Eye
ray-cast panorama, low resolution
Visual front end
insect-inspired engineered front end: colour-opponent and edge channels (Study 2)
Mushroom body
published model (Ardin et al.), one-shot target familiarity
Steering hub
ring attractor (after Sun et al.) or one of five rival arbiters with the same inputs and tuning budget; Study 2: evidence-gated ring, visibility channel for every arbiter
Avoidance
optic-flow balance and looming, added after the hub
Compass
sun compass; gyro yaw rate (Study 2)
Heading ring
insect-inspired model (EPG-style, this study): yaw rate moves the bump, sun corrects it (Study 2)
Path integrator
published model (Stone et al.), central-complex home vector
Flight control
cascade PID, motors, 6-DOF Crazyflie model
Odometry
body speed
Home view memory
insect-style nest learning flight, second mushroom body (Study 2)
Landing controller
engineered controller using the view memory: servo, descent (Study 2)
published insect model, reimplemented
insect-inspired component designed here
engineered part
added in Study 2
arbiter under test